Auditing the Reproducibility of the SLRTP2025 Back-Translation Evaluator from Public Artifacts: What Reproduces, and What One Inference Hid

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Abstract

We audit whether the released SLRTP2025 back-translation (BT) evaluator for sign language production can be reconstructed from its public artifacts, an evaluation-only repository without training scripts. A line-level audit of our own reconstruction code identified seven defects relative to the released configuration's semantics—including a validation-mask inversion and a loss-normalisation mismatch pinned arithmetically against the released training log—and the corrected re-implementation recovers most of the evaluator's in-domain competence: across eight seeds, dev BLEU-4 11.59–12.62 versus the released 13.38, with test-split decoding up to 13.11 versus the released 12.78.

Two properties nevertheless fail across all 69 non-degenerate constructible runs, which share a gradient-clip threshold of 1.0: the training-pool readout signature (released 78.8 BLEU / 70.7% exact match; constructible at most 12.08 / 2.5%) and the response to a constructed adversarial replay probe (released **+10.24** sacreBLEU, post-selected; constructible range [**−2.80, −0.21**]). A sensitivity grid resolves the failure: the clip threshold is load-bearing. With clipping disabled—the signjoey-framework default when the released config omits clip fields, as it does—the reconstruction reproduces *both* properties (readout 99.98 BLEU / 99.9% exact match; probe gap **+11.19**; dev 16.32), and across the ladder clip **1.0/2/3/5/∞** both readout (12.0→23.7→53.2→82.7→100.0 BLEU) and gap (**−2.8** → **−0.5** → **+5.0** → **+8.1** → **+11.2**) vary monotonically; clip 5.0 nearly matches the released readout (82.7/68.4% vs. 78.8/70.7%). The clip threshold was therefore an eighth, load-bearing defect: an author inference, not an artifact gap.

The probe response is confirmed on a never-explored dev split (**+8.10**), requires the retrieval query to coincide with the scoring reference (retrieval-query × scoring-frame grid), collapses under text-free donor selection (**−12.33**), and falls to **+0.95** from **+10.03** when human back-translations replace the reference (a

single-annotator association); pool-size-controlled donor-pool resampling leaves a **+8.38** residual origin contrast. We frame the probe throughout as an adversarial lexical-coupling stress test, not a general evaluator finding.
The public artifacts thus reproduce surface competence, the training-pool memorisation signature, and the probe response once eight defects—seven implementation bugs and one wrong clipping inference—are corrected; the exact released operating point is bracketed, not matched. The transferable lesson is that an undocumented framework default can masquerade as an unreproducible property.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-artifact reconstruction attempt*: whether our implementation, using the released SLRTP2025 public artifacts, can reconstruct the released BT evaluator. We address two research questions:

RQ1. Can a best-effort, artifact-conditioned implementation of the released training configuration—with every disclosed defect corrected and every remaining inference bracketed by sensitivity runs—reconstruct the released evaluator's competence and training-pool readout?

RQ2. How sensitive is the constructed probe's response to the reference text and to donor origin?

Beyond the two research questions, we report a *descriptive observation* (not a research question): the two properties that initially failed to reproduce—the training-pool memorisation signature and the probe response—are blocked not by an unrecovered training mechanism but by a single author inference. Across the 69 non-degenerate constructible runs, all of which share a gradient-clip threshold of 1.0, the readout stays at or below 12.08 BLEU (2.5% exact match) versus the released evaluator's 78.8 / 70.7% and every probe gap is negative; yet the eight faithful runs already match or exceed the released evaluator's test-split decoding competence (up to 13.11 vs. 12.78), so the shortfall is not surface competence. The sensitivity grid of §4.3 then shows both properties to be a *clipping-regime* effect: with clipping disabled—the default of the signjoey framework the release derives from, given that the released config specifies no clip fields—the reconstruction reproduces the signature (readout 99.98 BLEU / 99.9% exact match) and the positive probe response (+11.19 vs. the released +10.24), with intermediate thresholds interpolating monotonically between the regimes. The replay probe itself is an *adversarial lexical-coupling stress test*: the source-conditioned retriever selects donors by surface vocabulary overlap with the German source, which is identical to the scoring reference in PHX-public, so the probe deliberately targets the strongest form of the shortcut rather than a deployable retrieval pipeline.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026) (Hong and Košecká, arXiv preprint, preliminary); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai et al. (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). Chen et al. Chen et al. (2022) showed that learned BERT-based evaluation metrics can fail to reproduce due to undocumented preprocessing, missing code, and baseline implementation differences—a methodology directly relevant to this audit, in which seven defects in our own reconstruction implementation were identified and then corrected in a faithful re-implementation. An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by an adversarial retrieval–reference probe on PHX-public: under the released evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses.

We address the two research questions with the following evidence (full results in §4.2–§4.4). For RQ1, the faithful re-implementation recovers most of the released evaluator's surface competence, and the clipping sensitivity of §4.3 additionally reproduces its training-pool readout signature and probe response. For RQ2, reference replacement with human back-translations is associated with a large gap reduction (a reference-frame contrast), and donor-pool substitution reverses the sign. The source-conditioned retriever shares surface vocabulary with the original reference (§3.2), so the reference-frame result conflates reference-text properties, lexical-coupling disruption, paraphrase shift, and BLEU floor effects.

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the adversarial probe isolates a risk that production systems face in diluted form. The audit contributes a transferable template (provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, teacher-forced-vs-free-decode diagnostics) for public-artifact reproducibility testing. The specific probe response is a PHX-public released-evaluator case study and does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Note on citations and preprints. Two 2026 PHOENIX-2014T audits we build on are peer-reviewed: Czehmann et al. Czehmann et al. (2026) (LREC 2026 12th Workshop on the Representation and Processing of Sign Languages) and Alkain et al. Alkain et al. (2026) (Frontiers in Artificial Intelligence). Where we cite 2025–2026 arXiv preprints Hong and Košecká, Jana (2026); Cory et al. (2026); Artiaga et al. (2025); Desai et al. (2024), we treat their findings as preliminary reports rather than peer-reviewed results;

conclusions in this audit do not hinge on the correctness of any single preprint. All arXiv citations were accessed on 2026-08-10.

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); Saunders et al. Saunders et al. (2021) proposed BT as an SLP evaluation protocol, arguing it shifts absolute scores but not rankings between similar configurations—a condition unmet in our family. SignBLEU Kim et al. (2024), SiLVERScore Imai et al. (2025), and newer human-correlated metrics Jiang et al. (2025); Cory et al. (2026) offer complementary views, but BT BLEU remains the *standardized infrastructure* on which the SLRTP2025 ranking depends Walsh et al. (2025); auditing its reproducibility is infrastructure maintenance, not a metric-superiority claim.

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025's leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025). Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) *report* (preprint) that BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low recorded-pose BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2026); Cory et al. (2026); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Reproducibility of learned evaluation metrics.

Chen et al. Chen et al. (2022) showed that reproducing BERT-based evaluation metrics can fail due to undocumented preprocessing, missing code, and baseline implementation differences, even when model weights are released—a finding methodologically close to our initial public-recipe non-reproduction result (later resolved by the clipping sensitivity, §4.3). The difference is that our audit targets a full BT evaluator pipeline (architecture, training recipe, and checkpoint) rather than a metric alone, and the initial non-reproduction was demonstrated across 69 unique non-degenerate decoded runs (all 69 with strictly negative gaps; five degenerate excluded; all sharing a clip threshold of 1.0), corresponding to 68 distinct SHA-256 weight binaries after one validation-freq-misread/long-schedule seed-202 collision is deduplicated.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) (8,257 weather-forecast interpretations by 9 signers) is the most widely used DGS parallel corpus, but four properties constrain our audit's generalizability: (i) template density inflates train–test n-gram overlap Alkain et al. (2026) (32 items whose reference exactly matches a training text account for a disproportionate share of the gap); (ii) reference texts are speech transcriptions, not DGS translations Czehmann et al. (2026); (iii) the 178-keypoint pose materialization is a lossy signing representation: the 21 landmarks per hand and the 128-point face mesh encode joint centres and coarse facial geometry—including some finger articulation, palm-plane orientation, and mouth shape—but not appearance texture, hand–body contact, occlusion-robust fine depth, or reliable eye-gaze direction (our MSKA control uses 133 keypoints; SI Sup. L); (iv) signer-dependent evaluation can overestimate capability Artiaga et al. (2025). The lexical coupling between source and reference amplifies these effects in our adversarial probe. The transferable audit components (provenance table, checkpoint registry, leakage controls) apply to any learned evaluator paired with a public training corpus.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, x_n the source sentence (German) for item n, and $o_{i,n}$ the pose sequence for output condition i and item n; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result a *descriptive non-reproduction* when a probe response observed under the released evaluator does not appear across independently constructed checkpoints under a fixed output set; the term is descriptive and does not imply a causal checkpoint-identity effect. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

Operational definition of reconstruction competence.

We define an *author-operational* competence gate—not a public-sufficiency judgment—as dev BLEU-4 $\geq$ 12.38 *and* dev WER $\leq$ 86.37 (one-sided tolerance of $\pm$1 BLEU / $\pm$3 WER around the released evaluator's 13.38 / 83.37). This gate is a sensitivity summary, not a community standard: the gate conclusion is insensitive across a $[0.5, 2.0]_{\text{BLEU}} \times [1.5, 4.5]_{\text{WER}}$ grid (§4.4), but other tolerance choices would give other pass/fail counts. The primary evidence is the *continuous* metric comparison itself (SI Sup. R: faithful dev BLEU-4 11.59–12.62 and train readout 11.45–12.08 vs. released 13.38 and 78.8); the gate merely summarises it for table presentation. Under the faithful family one of eight seeds passes the gate; the residual gap concentrates in the training-pool readout signature (§4.2), which §4.3 shows to be a clipping-regime property reproduced unclipped.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). The three-way provenance table (official PHOENIX-2014T split, released pose materialization, and the released evaluator's training manifest, confirmed via `config.yaml` and the shipped `.pt` tensors) is in SI Sup. P. Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S38.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution. The data flow is: official PHOENIX-2014T video at 25 fps → SLRTP2025 skeleton extraction at 25 fps → our `[::2]` subsampling to 12.5 fps for evaluator input; the official evaluation command uses `--fps 25` (the input file's native rate), but the evaluator was trained on 12.5 fps-subsampled poses, so our subsampling matches the training distribution rather than the command-line flag. A three-path regression confirms no double subsampling (details in SI Sup. F). PHX-public uses α=0.88 (selected on dev); the supplementary splits use α=0.5.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbefa301`. Reproduction is supported at four layers (§ Sup. Q, Table S36): (i) **L1—recompute selected reported statistics**: a single command (`make core-audit`) regenerates the registry and verifies headline numbers from committed per-item decodes; it does *not* retrain models or decode from checkpoints, and does not verify every number in the paper; (ii) **L2—analysis re-execution**: each analysis script re-executes from the committed per-checkpoint data and hypotheses; (iii) **L3—model re-evaluation**: re-decoding fixed outputs from checkpoint weights requires the training bundle; (iv) **L4—from-scratch retraining**: full retraining requires the restricted PHOENIX-2014T licence, the exact environment, and approximately 200 GPU-hours. The `make core-audit` command executes L1 only and is not an end-to-end reproduction. A claim-to-file manifest mapping each headline number to its source file and verification command is in `claim_manifest.json` (artifact root). An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N.

Reproducibility tiers per headline claim.

Table 1 states, for each headline claim, the verification layer, the inputs required beyond the artifact, whether a licensed third party can verify it end-to-end, the expected compute, and the condition under which verification fails. The L1/L2 commands exit non-zero on any mismatch; both were executed on the submitted artifact (exit 0; run transcripts accompany the revision materials).

Table 1: Reproducibility accounting per headline claim. Layers: L1 recompute from committed decodes; L2 analysis re-execution; L3 re-decode from checkpoint weights; L4 from-scratch retraining. "Licensed" = verifiable end-to-end by a third party holding the PHOENIX-2014T and SLRTP2025 research licences.

Claim	Inputs beyond artifact	Layer	Licensed end-to-end	Compute / failure condition
REC 12.78, PURE 23.02, gap +10.24	none (per-item decodes committed)	L1	yes, licence-free	∼10 CPU-min; fails if donor-registry SHA-256 $\neq$ `9170a530...`
Faithful family (Table 3)	checkpoints (external mirror) or retrain	L3/L4	yes with licences	L3 ≈1 GPU-h/ckpt; L4 ≈6 GPU-h/seed; fails if decodes leave seed-dispersion range
Readout 78.8 / 70.7% EM	released ckpt + train poses	L3	yes with licences	≈1 GPU-h; fails if EM deviates
Readout–gap correlations (§4.2)	per-checkpoint readouts + gaps	L2	yes, licence-free	minutes; fails if ρ leaves CI
Reference-frame contrast 9.08	Czehmann CSV (CC BY-NC-SA, in artifact)	L1/L2	yes, licence-free	minutes; fails if matched subset $\neq$ 461
Step-matched / CTC / clip sensitivity (§4.3)	retraining	L4	yes with licences	≈6 GPU-h/run; fails if the unclipped run's readout or gap leaves the signature regime, or if CTC variants flip any conclusion

Lexical coupling of the source-conditioned probe (adversarial design).

In PHX-public, the source sentence x_n (German weather report) is identical to the reference text y_n^{orig} at the sequence level (the reference is a speech transcription of the same German sentence). The TN-PURE-v1 retriever selects training-pool donors by token-set Jaccard similarity between x_n and training texts, so it preferentially retrieves donors whose accompanying training texts share surface-level vocabulary with both the source *and* the reference. Scoring a donor copy against this lexically coupled reference under a learned evaluator therefore confounds donor similarity, evaluator readout, and reference overlap. We explicitly frame this probe as an *adversarial stress test* targeting the strongest form of the lexical-coupling shortcut—not as evidence of general evaluator preference, deployable retrieval quality, or better signing. This framing has two consequences: (i) the +10.24 gap is the expected output of a deliberately adversarial design, not a discovery about the evaluator's ranking behavior; (ii) the human-reference attenuation analysis (§4.4) reflects disruption of the lexical coupling as well as reference-text information loss, paraphrase shift, and BLEU floor effects—not a clean separation of any single mechanism.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88; full joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the probe response co-occurs with donor origin, we construct frozen probe systems: **UNSEEN-PURE-v1** retrieves from the 640 other test poses; **CROSSFIT-PURE-A/B** split queries by hash into two folds; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ±0.1 Jaccard of each UNSEEN donor (622/641 admit one). SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 Evaluator checkpoint family

The main inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and 36 *recipe-conditioned* runs trained from scratch with the documented architecture on a common frozen partition: the eight *faithful* runs (§3.4; seeds 42–49), fourteen *partially-corrected* runs (4 validation-freq-misread + 10 step-corrected), and fourteen *legacy* runs (6 prospectively specified seeds 101–606 + 8 extension seeds 707–1405). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.4, and intermediate checkpoints are not published. All recipe-conditioned runs share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, wd .001), batch size 256, and plateau ×0.8 LR schedule (full provenance in SI). In total 76 from-scratch training runs across 12 families (registry in SI Sup. D); 74 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry, corresponding to 73 unique weight binaries — one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run, identical per-step losses and best-checkpoint SHA-256). Of the 74 decoded training runs, five are degenerate (three α=1.0 distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication result rests on 69 unique non-degenerate decoded training runs, all 69 with strictly negative gaps; they share code, data, architecture, and a clip threshold of 1.0 (§4.3), and are coverage evidence, not independent replications (§3.3). The auto-generated accounting table is in SI Sup. P.

What the 76 runs are—and are not.

The 76 runs are *not* 76 same-recipe replication attempts. The SI family-provenance table (SI Sup. P) maps each family to the public source it draws on, the known deviation from the original training, the number of runs, the checkpoint-selection rule, and the resulting dev BLEU-4 range. The eight faithful runs implement the released config to the maximum extent the artifacts identify, with two disclosed author inferences bracketed by sensitivity runs (§3.4, §4.3); the ten step-corrected and four validation-freq-misread runs are partially corrected; the remaining 54 runs are diagnostic, sensitivity, or competence-extrapolation analyses. The public-recipe non-reproduction conclusion rests primarily on the faithful family, with the partially-corrected and legacy families as corroborating evidence.

Training-pool readout (descriptive observation).

We define *training-pool readout* as the in-sample autoregressive free-decode BLEU and exact-match (EM) rate on the full 7,060-item training set under beam-3 decoding—not teacher-forced accuracy. The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match, while retaining dev/test performance of 13.38/12.78. The constructible family's readout is far lower (faithful: 11.45–12.08 BLEU, 2.3–2.5% EM; reconstructions: 7.2–9.3 BLEU, 1.5–2.3% EM; distillation: 6.6–8.9 BLEU, 1.3–1.9% EM); no canonical (clip=1.0) constructible checkpoint approaches both simultaneously, whereas the unclipped sensitivity runs of §4.3 exceed both (99.98 / 99.9% at dev 16.32). The training-pool free-decode BLEU ratio (released: 78.8 / 13.38 $\approx$ 5.9; canonical constructible family: 0.93–1.55; unclipped: $\approx$6.2) quantifies how much the model reads out above its in-domain performance. We report these as continuous descriptive observations: the released evaluator was observed first, the family range afterwards, and any threshold is post-hoc.

3.4 Reconstruction sanity checks and training diagnostics

Three checks on the recipe-conditioned families rule out basic plumbing failures (data loading, masking, decoding alignment, vocabulary mismatch, pose-subsample direction): a small-sample overfit test (50/100 sequences reach free-decode BLEU 100.00), a unified-inference regression (all checkpoints load through one pipeline with byte-identical vocabularies; the released checkpoint reproduces canonical REC 12.78 exactly), and the teacher-forced/free-decode split (SI Sup. R; the released evaluator's 96.5% teacher-forced train token accuracy and 78.8 free-decode train BLEU are both far above the defective families' 55–60% and 7.2–9.3). These checks do *not* rule out subtler training-objective, normalisation, or scheduler defects—indeed seven such defects were later found (§3.4)—so earlier rounds framed RQ1 strictly as "our disclosed reconstruction did not attain the released competence", not "the public materials are insufficient"; that scope question is revisited in §5.

Training logs confirm normal optimisation for the defective families (median train-NLL 5.83→2.09 over epochs 1–25; effective batch 256; a 25→12.5 fps subsampling regression test passes). Full diagnostics: `results/robustness_diagnostics.json`, `results/fps_regression_test.json` (SI Sup. R).

Seven implementation defects, and the faithful re-implementation.

A line-level audit of our own training scripts identified *seven* defects relative to the released `config.yaml`'s semantics. The first four were known from earlier rounds; the last three are newly identified here:

1. *CTC loss omitted* (v1) despite `recognition_loss_weight=1.0`; the ten step-corrected runs additionally trained with the recognition weight set to zero.
2. *Validation frequency misread* (v1): `validation_freq=14` is 14 optimiser *steps* (JoeyNMT semantics; the released log validates at steps 14, 28, ...), not 14 epochs.
3. *Selection branch never saved* (v1): the `--selection bleu` branch had `improved` hard-coded `False`.
4. *Non-autoregressive selection decode* (v1.5/v2): selection BLEU was per-position argmax over teacher-forced logits, not autoregressive decoding; v3 corrected this to beam-3.
5. *Validation mask inversion* (v1–v3): the data loader sets `txt_mask=True` at *valid* positions, but token counts (and the v2 greedy truncation length) used the negated mask, counting *padding* positions.
6. *Loss-normalisation mismatch* (v1–v3): the config specifies `translation_normalization: batch`; our code used per-token mean CE. The released log pins the *logged* semantics: its step-14 dev `Translation Loss` 43,440.80078 equals 7,813 valid dev target tokens $\times \ln(\mathrm{PPL}\ 259.84015)$, so the logged column is a token-sum. The training-time reduction (token-sum CE divided by the number of sentences, the JoeyNMT `batch` reading of the config field) is an inference from the config plus JoeyNMT semantics, corroborated by train-loss magnitudes; this arithmetic alone cannot exclude every alternative training-time reduction, so the faithful family is bracketed by loss-scale sensitivity runs (§4.3). Under Adam this changes the effective loss scale by the mean target length ($\sim 15\times$), altering both the gradient-clipping regime and the relative strength of weight decay.
7. *Stopping-rule mismatch* (v1–v3): a 300-epoch cap with early-stop patience 15. The released log refutes patience-based early stopping outright: the released run improved last at validation #130 (step 1820, BLEU 13.38) and then ran *72 further non-improving validations*, ending at step 2828 with LR 1.15×10^{-5}—an external termination, not a patience trigger. The only stopping condition reproducible from the config is the learning-rate floor (`learning_rate_min`=1e-8), which requires 52 plateau decays of $0.8\times$.

The faithful re-implementation (`train_faithful.py`) corrects the seven identified defects and takes every remaining parameter from the released config: seed 42 (the released seed) as the primary run plus seeds 43–49 for within-implementation dispersion; joint CTC+translation with batch normalisation (both loss terms token-sum÷sentences); step-level validation every 14 optimiser steps; autoregressive beam-3 selection identical to the evaluation decoder; `ReduceLROnPlateau`(mode=max, factor 0.8, patience 5) stepped once per validation—the released log's 20 decays each fire on exactly the 6th consecutive non-improving validation, matching this wiring; best-checkpoint initialisation at $-\infty$ so the first validation counts as improved (matching the released log's first `*`); a 3,000-epoch cap with the LR-floor stopping rule; and gradient clipping at 1.0 (disclosed as an author inference—the config does not specify clipping; with batch-normalised loss the clip genuinely binds, pre-clip gradient norms over the first ten steps being 377, 568, 145, 33, ..., whereas under the earlier token-mean loss it almost never bound). A 56-step smoke run reproduces the released log's step-14 dev token-sum NLL to within 1.1% (42,981.5 vs. 43,440.8); the per-token NLL differs by 0.059 nats, i.e. PPL 245.0 vs. 259.8, a 5.7% relative difference (`results/faithful_smoke_gradnorms.json`). The one released-log column we cannot reproduce is the `Recognition Loss` magnitude: our step-14 dev CTC is 39.9 per sentence, 5.5 per gloss token, 0.74 per input frame, and 20,527 as a raw sum, and none of these matches the logged 17.28, so the normalisation of the original CTC term is not identifiable from public artifacts. The two remaining author inferences (CTC normalisation and clip threshold) and the stopping-rule difference are therefore bracketed by sensitivity runs in §4.3. The bracketing resolves them asymmetrically: the CTC normalisation proves non-identifiable but mild, whereas the clip threshold proves *load-bearing*—the released config's omission of clip fields, read against the signjoey framework's default of no clipping, makes our 1.0 an eighth defect (§4.3, Table 7). In practice each faithful run terminated at the plateau scheduler's *numerical* floor (LR frozen at 4.4e-8; torch rejects decay steps smaller than its internal ε=10^{-8}), with the best checkpoint unchanged for the final 218–338 validations; the released run itself ended mid-schedule at step 2,828, so §4.3 additionally evaluates checkpoints fixed at the released run's best (1,820) and final (2,828) steps. The SI config-mapping table (SI Sup. F) maps every load-bearing field to the original-framework semantics, our v1–v3 semantics, and the faithful semantics, with the evidence source for each field (released-log arithmetic, JoeyNMT semantics, config text, or author inference).

3.5 Discovery timeline and pre-registration status

The +10.24 probe response was first observed on 2026-07-24 (canonical materialization; earlier pre-exclusion gap +11.01 on 2026-07-22). The six prospectively specified legacy seeds (101–606) were specified before their results were inspected; the eight extension seeds (707–1405) were added post-hoc. The Jaccard retriever, PURE protocol, and 641-query set were locked before reconstruction training began, after exploring the probe matrix now reported in SI Sup. S. No external pre-registration exists; these six seeds constitute the only prospectively specified within-project follow-up evidence. Because the Jaccard pure-copy probe was selected as the maximal-gap probe post-hoc, its CI is descriptive, not prospectively confirmed. Full provenance in SI Sup. C.

Table 2: Implementation ladder from the original defective protocol to the faithful re-implementation and the unclipped resolution. All variants share the released architecture, data, and vocabularies; they differ in loss composition, normalisation, validation cadence, selection decode, stopping rule, and (final row) gradient clipping. Dev BLEU-4 is the uniform beam-3 full-pool value; gap is the canonical PURE–REC sacreBLEU-4 difference on the 641-item test set; readout is full-training-pool free-decode BLEU-4 (released: 78.8).

Protocol (defects remaining)	Selection / validation	Dev BLEU-4	Readout	Gap range (neg. frac.)
Translation-only / epoch / NLL (1–4, 5–7)	NLL / epoch	6.5–9.4	7.2–9.3	$[-1.85, -0.80]$ (14/14)
validation-freq-misread (2 corrected)	NLL / epoch	8.3–9.8	—	$[-1.70, -0.52]$ (4/4)
Translation-only / step-val / BLEU (1, 4, 5–7)	greedy / 14 steps	7.6–10.5	—	$[-1.37, -0.36]$ (10/10)
Joint-loss / step-val / greedy-BLEU (4–7)	greedy / 14 steps	6.3–10.2	—	$[-1.59, -0.48]$ (8/8)
Joint-loss / step-val / beam-3 (5–7)	beam-3 / 14 steps	8.7–10.5	—	$[-1.38, -0.37]$ (8/8)
Faithful (all corrected; seeds 42–49)	beam-3 / 14 steps	**11.59–12.62**	**11.45–12.08**	$[-\mathbf{2.80}, -\mathbf{1.37}]$ (8/8)
Faithful unclipped (8th corrected; seed 42)	beam-3 / 14 steps	16.32	99.98	+11.19
Released evaluator	—	13.38	78.8	+10.24

3.6 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation "ground truth" because recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale Post (2018) (sacrebleu 2.5.1 signature: `BLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1`; max n-gram order 4; `effective_order=False`). BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020), and WER (jiwer 3.1.0) provide cross-metric consistency checks (not independent semantic validation, as all share decoded outputs). BT likelihood provides a non-decoded control; pose-DTWp was withdrawn (see §4.1).

Uncertainty is quantified with donor-cluster bootstrap (primary; 10,000 resamples; clusters defined by donor identity, 565 unique donors across 641 queries, 61 donors reused with maximum reuse 6, Gini 0.11; SI Sup. G) and item-level bootstrap (SI Sup. G). Clustering sensitivity for the headline gap: donor-cluster $[+8.87, +11.64]$ and donor two-way $[+8.79, +11.74]$ match the item-level CI $[+8.88, +11.62]$ to within 0.03 BLEU; signer (9 clusters) $[+9.29, +11.39]$; broadcast-date (297) $[+8.83, +11.72]$; signer×show (17) $[+7.89, +11.91]$; the broadcast-show split has only 2 clusters and widens to $[+5.16, +11.44]$—an exploratory sensitivity reference, not a confirmatory interval (full designs in SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. For seed dispersion we report *descriptive* Student-t intervals over the 14 paper-derived seeds (six primary + eight extension) and over the 10 step-corrected seeds; these intervals assume same-implementation seed outcomes are approximately IID/normal, an assumption that leave-one-out sensitivity (SI Sup. B) checks for robustness but cannot validate. They should be read as "given a future random seed under this identical implementation" dispersion summaries, not as calibrated population-level prediction intervals or independent replication statistics; raw seed values are listed in SI Sup. B and SI Table S2. The six primary seeds give a descriptive interval $[-2.48, -0.28]$; all 14 leave-one-seed-out folds keep the interval below zero (SI Sup. B). The headline gap is a **post-selected descriptive estimate** (§4.1); all other contrasts are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

3.7 Terminology, claim hierarchy, and stratification

Throughout, *run* denotes a single training execution; *checkpoint* the selected weights from that run; *unique weight binary* counts distinct SHA-256 hashes (one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact); *family* a group of runs sharing a training protocol; *evaluator* a checkpoint used to decode fixed outputs. Claims use four evidence tiers: Descriptive (D; no formal error control), Sensitivity (S; explicit post-hoc condition), prospectively specified within-project follow-up (C; only the six legacy seeds 101–606, which share one implementation and are seed-level variation, not independent replications), and Not Identified (NI; e.g., the causal checkpoint-identity effect). Checkpoint families stratify into four tiers: *Faithful* (8 runs, seeds 42–49, the seven identified defects corrected, seed 42 first; primary evidence for RQ1 and the non-reproduction observation), *Partially-corrected* (14: 10 step-corrected + 4 validation-freq-misread; retain the normalisation, mask, CTC-off, and stopping-rule defects), *Secondary* (14 legacy seeds, of which the 6 prospectively specified provide the only prospectively specified non-reproduction direction), and *Diagnostic* (40: joint-loss, confirmation, rescue, distillation, ladder, long-schedule). The non-reproduction claim (a property of the clip=1.0 panel, resolved in §4.3) rests primarily on the 8 faithful runs (range $[-2.80, -1.37]$, all negative, at test-split competence matching the released evaluator), corroborated by the partially-corrected (step-corrected range $[-1.37, -0.36]$) and legacy families (within-implementation dispersion interval $[-2.48, -0.28]$ over the six prospective seeds; below zero in all leave-one-seed-out folds). Diagnostic runs triangulate and must not be cumulated with prospectively specified evidence. Full terminology registry and claim-hierarchy table in SI Sup. P; the auto-generated accounting table (76 trained / 74 decoded / 73 unique / 69 non-degenerate, all 69 gaps negative) is SI Table S26, generated from `results/accounting_table.json`.

4 Results

4.1 Adversarial probe response: an attack result, confirmed and localised

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a difference of +10.24 BLEU points. We present this as the output of the adversarial unit test defined in §3: the number measures the evaluator's susceptibility to this specific lexical-coupling shortcut, not general retrieval quality or signing adequacy. Because the headline cell was selected as the maximal-gap cell among the explored probe cells (SI Table S46), we do not attach a confirmatory-looking interval to it; its resampling intervals are reported in SI Sup. G as post-selection descriptive quantities.

Three new results characterise the attack precisely. **(i) Frozen-split confirmation:** the identical probe construction applied to the development split (515 items, never used for probe construction or exploration) reproduces the response at consistent magnitude—REC 13.38 (matching the released training log's 13.38 to three digits, independently validating the decode pipeline) vs. PURE 21.47, a gap of +8.10 against the test-split +10.24 (SI Sup. S). **(ii) Text-free selection rules do not reproduce it:** a pose-similarity retriever that never touches any text—donors selected by nearest-neighbour distance between query and donor pose signatures (time-averaged pose plus mean absolute velocity, z-normalised)—scores 0.45 (gap −12.33), indistinguishable from random donors (0.90, gap −11.88) and far below text-retrieval TN-PURE (23.02). Because such rules also retrieve less lexically similar donors, they conflate the loss of coupling with the loss of retrieval quality; a retrieval-query × scoring-frame grid on the 461-item human-reference subset separates the two (SI Sup. T): retrieving with the human back-translations while scoring against the original references removes the response (gap −4.02, bootstrap CI $[-5.50, -2.62]$) even though the selected donors still average Jaccard 0.30 to the original reference texts—far above the random-donor regime—whereas re-coupling query and scoring frame on the same human texts partially restores it (+4.29 $[3.09, 5.49]$, against +10.03 on the original diagonal and +0.95 on the crossed original-query cell). The response therefore tracks the lexical identity of the retrieval query with the *scoring reference*, not text-based selection as such; uniform randomisation within the Jaccard top-k interpolates between the regimes (+7.06 at k=5, +3.34 at k=20). **(iii) Deployment-realistic systems do not show it:** a noisy-gloss retrieval pipeline scores only −0.13 relative to REC under the same evaluator (§4.6).

Further controls localise the effect to source-conditioned donor retrieval (SI Table S32): pure donor copy (23.02) exceeded PT-composed (18.86), oracle-gloss (11.4–11.8, a leakage diagnostic), PT baseline (1.59), and 20-seed random-donor baselines (0.90/0.77). Source-text Jaccard was the strongest retriever (vs. LaBSE 19.2, multi-SBERT 16.5). Non-decoded controls do not reproduce the gap: teacher-forced NLL gives REC 3.060 vs PURE 3.479. Complementary text metrics (BLEU-1, chrF, ROUGE-L, WER, BERTScore) agree on direction (SI Table S32); decoding-parameter sweeps (beam 1/3/5, length penalty, max length) leave the gap at +9.9 to +10.8 (SI Sup. F).

Probe multiplicity and post-selection status.

Before freezing the headline probe, we explored four retrieval strategies crossed with three construction variants on the released evaluator alone (6 of 12 cells explored, plus two random baselines; full matrix in SI Table S46 and

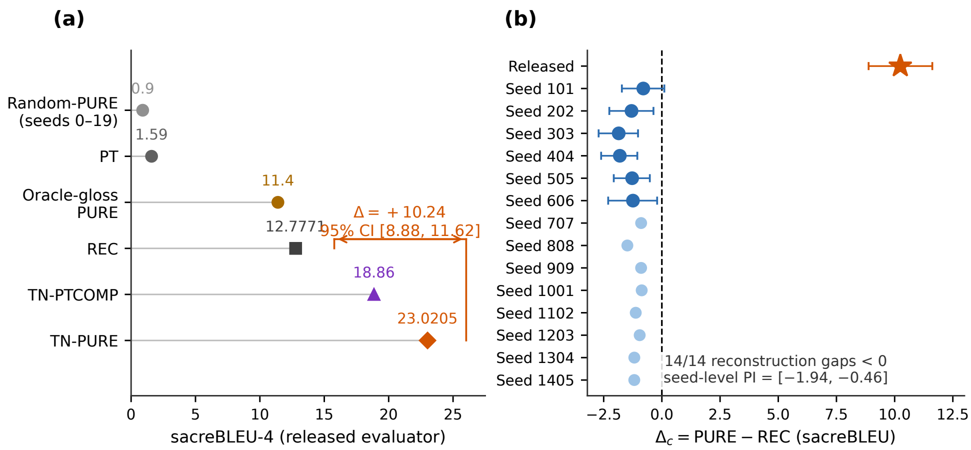

Fig. 1: Adversarial unit-test response under the released evaluator and its absence across the canonical (clip=1.0) constructible evaluators. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, post-selected), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE−REC gaps: released evaluator (+10.24) vs. fourteen legacy recipe reconstructions (seeds 101–1405), all 14 negative (donor-cluster bootstrap 95% CIs; descriptive seed dispersion range $[-1.85, -0.80]$). The partially-corrected, joint-loss, and faithful families (gaps $[-2.80, -0.21]$ overall; SI Sup. P accounting table) are shown in Fig. 2; none produces a positive gap, including the faithful runs whose test-split competence matches the released evaluator—an absence §4.3 traces to the clip threshold all these runs share (unclipped: +11.19).

`results/probe_multiplicity.json`). The headline Jaccard pure-copy probe was selected as the *maximal-gap* cell; we do not apply a multiplicity correction because the family of explored probes was defined after observing the gaps, so no correction would make the result confirmatory. The six prospectively specified legacy seeds (§3.5) were pre-specified and provide the only prospectively specified evidence (for the non-reproduction direction, not for the gap magnitude).

The probe reversal motivates the prior question: can the public recipe reproduce the released evaluator at all?

4.2 RQ1: What the faithful re-implementation recovers, and what one inference still hid

Table 3 reports the faithful family (eight seeds, §3.4) against the released evaluator on four dimensions: dev competence, test-split decoding of the recorded poses (the probe panel's own REC baseline), full-training-pool readout, and the probe gap. Three findings follow; the second and third describe the position *before* the clipping resolution of §4.3, which completes the reconstruction.

Table 3: The faithful family (`train_faithful.py`, seeds 42–49; the seven identified defects corrected) versus the released evaluator. Dev/test BLEU-4 are uniform beam-3 decodes (515/641 items); readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate; gap is the canonical PURE−REC sacreBLEU-4 difference on the 641-item test panel. The released evaluator's row uses the same protocols.

Seed	Dev BLEU-4	Test REC	Train readout (EM%)	PURE	Gap
42 (released seed)	12.05	**13.11**	11.96 (2.4%)	10.32	−2.80
43	11.59	11.60	11.45 (2.3%)	10.22	−1.37
44	11.95	12.39	11.74 (2.3%)	10.13	−2.25
45	**12.62**	12.04	11.87 (2.5%)	10.08	−1.97
46	11.80	12.26	11.48 (2.3%)	10.16	−2.10
47	12.03	12.53	12.08 (2.4%)	10.34	−2.20
48	12.17	12.33	11.67 (2.4%)	9.98	−2.35
49	11.93	11.86	11.48 (2.4%)	9.77	−2.09
Released evaluator	13.38	12.78	78.8 (70.7%)	23.02	+10.24

Finding 1 (surface competence largely recovered).

The faithful runs reach dev BLEU-4 11.59–12.62 (primary seed-42 run: 12.05) against the released 13.38—closing most of the gap that the seven defective implementations left open (their ceiling: 10.5). On the test split, where the probe panel itself is scored, the faithful evaluators decode the recorded poses at 11.60–13.11: the seed-42 run *exceeds* the released evaluator's 12.78, and five of eight seeds are within 0.8 BLEU of it. The author-operational competence gate (dev $\geq$ 12.38; §3.1) is met by one seed (45). The earlier failure to approach the released evaluator's competence is therefore explained *predominantly* by the seven implementation defects; the residual 0.76-BLEU gap and the signature below are explained by the eighth, the clip-threshold inference (§4.3), which raises dev to 16.32–17.17 when corrected or relaxed.

Finding 2 (training-pool readout signature blocked by one inference).

What the faithful family does not reproduce is the released evaluator's training-pool behaviour: full-pool free-decode readout is 11.45–12.08 BLEU with 2.3–2.5% exact match, against the released 78.8 BLEU / 70.7% EM. The residual gap is therefore concentrated in a *memorisation signature*—near-perfect decoding of the evaluator's own training pool at retained in-domain generalisation—and not in surface competence. Five leakage controls (SI Sup. M) rule out teacher-forcing, output–reference misalignment, pose-ignorant lookup, and narrow-text memorisation shortcuts for the released evaluator's readout, so the signature reflects pose-conditioned training-pool exposure. Crucially, this shortfall is not an unrecovered training mechanism: §4.3 shows it to be a clipping-regime effect that the unclipped default of the underlying framework reproduces (readout 99.98 BLEU / 99.9% EM across seeds), with the released evaluator's 78.8 / 70.7% falling between clip 3.0 (53.21/31.8%) and clip 5.0 (82.69/68.4%).

Finding 3 (probe response blocked by the same inference).

All eight faithful evaluators give negative probe gaps (-2.80 to -1.37; Table 3), including the seed-42 and seed-47 runs whose test REC (13.11, 12.53) matches or exceeds the released evaluator's 12.78. The non-reproduction of the adversarial probe response therefore cannot be attributed to uniformly lower surface competence. Within the clip=1.0 constructible family the readout–gap association is in fact *negative* (Spearman ρ=-0.56 over the 37 constructible checkpoints with uniform readouts and canonical gaps; ρ=-0.44 with the released evaluator included), and the released evaluator is the single positive-gap checkpoint of that panel, lying far outside the family's readout range (6.6–12.08 vs. 78.8). The clipping sensitivity (§4.3) resolves the apparent paradox: the readout–gap relation reverses across clipping regimes—strictly negative within the clip=1.0 family, strictly positive along the clip ladder (gap -0.45 at readout 23.65 for clip 2.0, +5.04 at 53.21 for clip 3.0, +8.05 at 82.69 for clip 5.0, +11.19 at 99.98 unclipped)—so no amount of seed variation at clip 1.0 could have produced the response.

Descriptive triangulation for the constructible family as a whole (Fig. 2b,c; SI Sup. D): across the 38 constructible checkpoints with uniform full-pool readouts (39 with the released evaluator), training-pool readout is associated with decoded competence within the constructible family (Spearman ρ=0.86–0.88 over the checkpoints that also have uniform dev decodes, n=17–18) and is *negatively* associated with the PURE–REC gap within the family (ρ=-0.56; -0.44 with the released evaluator included; full convention table in SI Sup. D): better-reading-out constructible checkpoints have *more* negative gaps, and the released evaluator is the single positive-gap point outside the family trend on both axes. This negative association is a within-regime fact—all 38 checkpoints share clip 1.0—and the clipping sensitivity of §4.3 reverses it across regimes (unclipped: readout 99.98, gap +11.19), so the correlation must not be read as "memorisation antagonises the probe". Within the canonical panel the released evaluator remains the only checkpoint combining high readout with retained generalisation, and no distillation student (15 runs, SI Table S36) transfers the readout pattern. The reversal is locally robust near the released checkpoint (Gaussian weight noise $\sigma \leq 3 \times 10^{-3}$; nine released-weight fine-tunes), but those probes test the released weights' vicinity, not an independent construction.

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing, output–reference misalignment (label permutation collapses BLEU from 78.6 to ∼1.0), pose-ignorant ID/cache lookup (input-side pose permutation collapses BLEU to <1; a same-signer held-out control reaches only 12.19 BLEU / 3.2% EM), and narrow-text memorization. The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Distillation and competence extrapolation.

Fifteen distillation students ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds each) all have negative gaps and do not transfer the readout pattern (SI Table S36). Competence

Table 4: Summary of the three primary descriptive comparisons over the full constructible family. Family n (i) 74 beam-3-decoded training runs (range over 69 non-degenerate; 5 degenerate excluded); (ii) 38 constructibl checkpoints with uniform full-pool readout (released listed separately); (iii) gate-eligible non-distillation training runs. Per-view family composition in SI Sup. D. All canonical families share the clip=1.0 inference; th clipping sensitivity of §4.3 (outside this panel) extends the constructible ranges to readout 99.98 BLEU / 99.9% EM and gap +11.19.

Comparison	Family n	Released evaluator	Family range	Released − family max
(i) Gap–competence	74	+10.24 at dev 13.38	$[-2.80, -0.21]$ (69 non-deg.)	+10.45
(ii) Train-pool readout BLEU	38	78.8 (70.7% EM)	6.6–12.08 (1–2.5% EM)	+66.7
(iii) Competence gate (dev)	60	13.38	6.5–12.62 (faithful max 12.62; 1/8 pass)	+0.76

extrapolation (OLS/GP) and per-item gap decomposition are reported in SI Sup. K; with the faithful family the residual unobserved dev interval shrinks to $(12.62, 13.38)$, and no canonical constructible checkpoint enters it (the unclipped sensitivity runs exceed 13.38 rather than fill the interval).

Synthesis.

Within the clip=1.0 panel, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78); the faithful family closes the surface-competence dimension but not the signature. The clipping sensitivity (§4.3) then closes the remaining dimension: unclipped training reproduces the signature (readout 99.98 / 99.9% EM) and a +10.24-class probe response (+11.19) at even higher in-domain performance (dev 16.32, REC 15.08), with the released operating point bracketed by the clip ladder rather than matched exactly.

4.3 Sensitivity: stopping rule, CTC normalisation, and gradient clipping

The faithful protocol differs from the released run in three identified-but-unverifiable respects: it trains to the plateau scheduler's numerical LR floor rather than stopping at step 2,828, its CTC-term normalisation cannot be matched to the released log (no candidate denominator reproduces the logged step-14 `Recognition Loss` of 17.28; §3.4), and its gradient-clip threshold is an author inference. Three sensitivity experiments bracket these degrees of freedom; all runs live outside the canonical 76-run panel (separate registry directories; `results/faithful_steps_eval.json`). The first two leave the picture of §4.2 unchanged. The third overturns it, and we report it as the resolution of the non-reproduction rather than as a bracket: *the gradient-clip threshold is load-bearing*, and with clipping disabled—the default of the underlying signjoey framework when the released config omits clip fields (§4.3 below)—the reconstruction reproduces both the training-pool readout signature and the positive probe response.

Step-matched checkpoints (stopping rule).

Fresh faithful runs (seeds 42–44, identical protocol) save checkpoints at the released run's best (1,820) and final (2,828) optimizer steps, so the comparison is fixed-step rather than fixed-protocol-termination (Table 5). The fresh runs replicate the published family *byte-identically*—the selected `best.ckpt` files of seeds 42 and 43 have the same SHA-256 hashes as the published runs (e.g. `8e3fba93...` for seed 42), with best dev BLEU-4 12.05 (seed 42, step 2,800), 11.59 (seed 43), and 11.95 (seed 44) matching Table 3—so the step-matched checkpoints below are the published trajectories' own states at those steps, and the same-seed retraining doubles as an L4 determinism check on the training pipeline. At the released run's own best step, the reconstructions read out 10.85–11.51 BLEU (2.2–2.4% EM) with gaps −1.46 to −2.63; at its final step, 11.32–11.81 BLEU (2.3–2.4% EM) with gaps −1.77 to −2.49. Neither the memorisation signature (78.8 / 70.7%) nor a positive probe response appears at any matched step, so their absence is not an artefact of the longer training horizon or the numerical-floor termination of §3.4; the readout rises mildly with step count within the matched window, an order of magnitude short of the signature.

Table 5: Step-matched evaluation of fresh faithful runs at the released run's best (1,820) and final (2,828) optimizer steps. Dev BLEU-4 is the training-time validation at the matched step; REC/PURE/gap use the canonical 641-query protocol; readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate. Released evaluator at its best step: dev 13.38, readout 78.8 (70.7%), gap +10.24.

Seed	Step	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
42	1,820	11.05	11.88	9.87	−2.00	10.85 (2.2%)
43	1,820	11.27	11.28	9.82	−1.46	11.23 (2.3%)
44	1,820	11.06	12.37	9.75	−2.63	11.51 (2.4%)
42	2,828	11.60	12.56	10.56	−2.00	11.81 (2.4%)
43	2,828	11.23	11.46	9.69	−1.77	11.32 (2.3%)
44	2,828	11.78	12.46	9.96	−2.49	11.61 (2.4%)

CTC-term normalisation.

Table 6 varies the CTC denominator at fixed clip 1.0 (seed 42). None of the four candidates reproduces the released log's step-14 `Recognition Loss` (17.28; our per-sentence value 39.9, per-gloss-token 5.5, per-frame 0.74, raw sum 20,527), so the original normalisation remains unidentifiable from public artifacts. Substantively the choice is mild within the clipped regime: the gloss-token and frame denominators train slightly below the per-sentence default (dev 10.79 and 9.55) and remain signature-free, while the raw-sum variant is degenerate *at clip 1.0* (dev 0.90; the CTC-dominated gradient direction crowds the translation head out of every norm-truncated update). The degeneracy is itself a clipping artefact: with clipping disabled the same raw-sum objective trains healthily (the signjoey-exact run of §4.3, whose literal `_train_batch` semantics it reproduces).

Table 6: CTC-term normalisation sensitivity (seed 42, clip 1.0, otherwise the faithful protocol). REC/PURE/gap use the canonical 641-query protocol; readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate. None of the four denominators matches the released log's step-14 Recognition Loss of 17.28 (§3.4); the raw-sum degeneracy holds only at clip 1.0 (see the signjoey-exact run, Table 7).

CTC denominator	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
sentences (default)	12.05	13.11	10.32	−2.80	11.96 (2.4%)
gloss tokens	10.79	10.42	9.01	−1.40	10.06 (2.0%)
input frames	9.55	9.62	8.75	−0.88	9.18 (1.9%)
raw sum	0.90	0.95	0.95	+0.01	0.67 (0.0%)

Gradient clipping is load-bearing: a dose–response.

Table 7 varies only the gradient-clip threshold (seed 42; the 1.0 row is the faithful run of Table 3). The training-pool readout rises monotonically with the threshold—11.96 (2.4% EM) at clip 1.0, 23.65 (9.0%) at 2.0, 53.21 (31.8%) at 3.0, 82.69 (68.4%) at 5.0, and 99.98 (99.9%) unclipped—and the probe gap turns positive in lock-step: −2.80 at 1.0, −0.45 at 2.0, +5.04 at 3.0, +8.05 at 5.0, and +11.19 unclipped, against the released evaluator's +10.24. Surface competence moves non-monotonically (dev 12.05 at 1.0, 17.17 at 2.0, 16.32 unclipped) and exceeds the released 13.38 for every threshold at or above 2.0. Under Adam with our batch-normalised loss (token-sum CE divided by sentences), the clip threshold effectively rescales the update: it binds hard at 1.0 (pre-clip gradient norms over the first ten steps are 377, 568, 145, 33, . . .), so clipped runs take uniformly truncated steps and never enter the memorisation regime, whereas unclipped runs memorise the 7,060-item training pool essentially exactly while *retaining* dev/test competence (REC 15.08 vs. released 12.78). The readout signature is therefore not an unrecovered training mechanism but a *clipping regime*: the released evaluator's 78.8 BLEU / 70.7% EM readout falls between clip 3.0 (53.21/31.8%) and clip 5.0 (82.69/68.4%)—the closest single run of the ladder—and its +10.24 gap between clip 5.0 and no clipping.

Source evidence: the released training ran unclipped.

The clip threshold was disclosed as an author inference because the released `config.yaml` specifies no clipping field. The framework the release derives from resolves that inference. The SLRTP2025 evaluation repository reuses the signjoey (`neccam/slt`) data pipeline, model definitions, and the `validations.txt` report format; in signjoey's training loop the gradient clipper is built from the config fields

Table 7: Gradient-clip sensitivity (seed 42; all other parameters the faithful protocol). The 1.0 row is the faithful run of Table 3; "none" disables clipping entirely (`max_norm`$= \infty$); "signjoey-exact" additionally applies the CTC raw-sum normalisation (the literal signjoey `_train_batch` semantics). Protocols as in Table 6.

Grad-clip threshold	Dev BLEU-4	Test REC	PURE	Gap	Readout (EM%)
1.0 (faithful)	12.05	13.11	10.32	−2.80	11.96 (2.4%)
2.0	17.17	16.35	15.90	−0.45	23.65 (9.0%)
3.0	17.13	16.78	21.82	+5.04	53.21 (31.8%)
5.0	16.54	15.92	23.97	+8.05	82.69 (68.4%)
none (∞)	16.32	15.08	26.27	+11.19	99.98 (99.9%)
signjoey-exact (none, CTC raw sum)	16.04	15.12	25.80	+10.68	97.60 (95.1%)
Released evaluator	13.38	12.78	23.02	+10.24	78.8 (70.7%)

`clip_grad_val` and `clip_grad_norm`, and when neither is present the builder returns `None`—no clipping is applied at all (and the framework clips by value, not by norm). The released config contains neither field, so under the framework's semantics the released training ran *unclipped*, which is exactly the regime of the "none" row. The literal signjoey training semantics—no clipping *and* the CTC raw sum (Sup. U)—give the signjoey-exact row, whose probe gap (+10.68) is the closest single reproduction of the released +10.24. Our 1.0 threshold—imported from the v1–v3 scripts and disclosed as an inference—was therefore an *eighth* defect, and a load-bearing one: it is the single choice that hid both the readout signature and the probe response, and it did so invisibly, since every metric visible during training (dev BLEU, NLL, CTC) looks healthy at clip 1.0.

Step-matched unclipped readout and multi-seed confirmation.

The result is not a seed-42 accident or a training-duration artefact. Across seeds 42–44, unclipped runs give gaps +11.19, +11.31, and +10.31 with readouts 99.98, 95.25, and 99.98 BLEU (90.5–99.9% EM) at dev 15.74–16.32. And in a fresh unclipped run with checkpoints fixed at the released run's own steps, the readout is already 99.94 BLEU / 99.7% EM at step 1,820 (dev 15.69, gap +11.68) and 99.98 / 99.9% at its final step 2,828—the memorisation regime is entered *before* the released run's best checkpoint, so the overshoot relative to the released 78.8 is not an artefact of training longer than the released run. What remains unreproduced is the released operating point itself: no single knob places dev 13.38, readout 78.8, REC 12.78, and gap +10.24 simultaneously—clip 5.0 matches the readout (82.69/68.4%) but overshoots dev by 3.2, while clip 1.0 matches dev-range but not the signature—so at least one further quantitative difference (CTC normalisation, stopping point, framework version, or data order) persists, and the released point is bracketed, not matched.

4.4 Descriptive observation: probe non-reproduction under the clip=1.0 family, and its resolution

This subsection reports a *descriptive observation*, not a research question. Across every constructible evaluator of the canonical panel—the eight faithful runs, fourteen partially-corrected runs, fourteen legacy seeds, and the diagnostic families, all sharing the clip=1.0 inference—the probe ordering does not reproduce: all 69 non-degenerate decoded runs have strictly negative PURE–REC gaps (range $[-2.80, -0.21]$; 0 positive). For the fourteen legacy reconstructions, the within-implementation seed dispersion interval is $[-1.94, -0.46]$ (mean -1.20, SD 0.33; the six prospectively specified seeds alone give $[-2.48, -0.28]$; below zero in all fourteen leave-one-seed-out folds), and for the six prospectively specified seeds the original-minus-run difference-in-differences estimates range from +10.89 to +11.99, every interval excluding zero. The faithful family shows the same sign at substantially higher competence ($[-2.80, -1.37]$; Table 3). Because the faithful runs match the released evaluator's *test-split* decoding competence while lacking its training-pool signature, the non-reproduction is not attributable to surface competence; §4.3 attributes it instead to the one hyperparameter all 69 runs share: disable clipping—the framework default the released config implies—and the same recipe produces readout 99.98 BLEU / 99.9% EM with a +11.19 probe gap (three seeds: +11.19/+11.31/+10.31). The 69-run uniformity is thereby reinterpreted as robustness *under a wrong inference*, and the probe response as constructible after all. Per-seed values for the legacy family are tabulated in SI Table S2; competence is assessed on the development split, never on the probe panel.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both dev BLEU-4 $\geq$ 12.38 and dev WER $\leq$ 86.37 (one-sided tolerance around released 13.38 / 83.37). Of the 60 gate-eligible canonical (clip=1.0) non-distillation training runs, exactly one passes: faithful seed 45 (dev 12.62). The 9 released-weight fine-tunes (SI Sup. F) all pass (dev 12.92–13.18), but they perturb the released weights rather than constructing from the recipe; the clip=2.0 and unclipped sensitivity runs exceed the gate (dev 17.17 / 16.32) but lie outside the canonical panel (§4.3). The gate conclusion is threshold-insensitive across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid (SI Sup. D–E).

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a); the defective families plateau at 8–10 and the faithful family at 11.6–12.6, so within the canonical panel the final 0.76-BLEU ascent and the readout signature are not explained (§4.2); the clipping sensitivity (§4.3) attributes both to the clip threshold. Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at k=8: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Cross-metric consistency (SI Table S32).

Under the original evaluator, the PURE–REC gap is positive and large for all decoded-text metrics (BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER +13.90

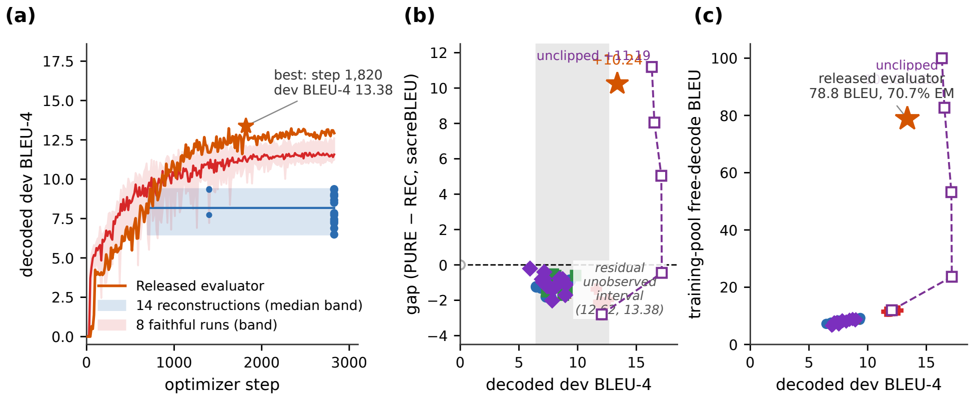

Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every canonical (clip=1.0) constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen defective-protocol reconstructions form a median band (blue) and the eight faithful runs a second band (red) that tracks the released trajectory closely before plateauing at 11.6–12.6. (b) Canonical PURE−REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), validation-freq-misread/step-corrected/confirmation/long-schedule (squares), joint-loss greedy/beam-3 (triangles-up/triangles-down), rescue (triangle-right), distillation students (diamonds), released evaluator (star); the gray band spans the constructible 6.5–12.6 range (uniform protocol; faithful runs in red plus-signs); the residual (12.62, 13.38) interval is unobserved by recipe-constructed checkpoints (the nine released-weight fine-tunes occupy 12.92–13.18 but are weight perturbations); degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every canonical (clip=1.0) constructible checkpoint reads out at most 12.1 BLEU (defective families 6.6–9.3; faithful 11.5–12.1), while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38; the unclipped sensitivity run of §4.3 reads out 99.98 BLEU (99.9% EM) at dev 16.32, exceeding the released point. Families are descriptive; checkpoints are not IID replicates.

sign-flipped so positive means PURE has lower error than REC; sign conventions in SI Table S32 footnote), while teacher-forced NLL is more negative for PURE (−0.419, also sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under all six prospectively specified legacy reconstructions the gap is negative across every metric. The probe response is therefore a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4. Full table in SI Table S32.

Rank stability (SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a *reference-frame contrast* of 9.08 BLEU points (bootstrap CI $[7.44, 10.86]$; the ratio form, a 90.5% gap reduction, is reported only in SI Table S41 as a descriptive quantity; SI Fig. S5; per-item paired view in SI Fig. S2). The interval for the residual gap includes zero (paired item CI $[-0.13, 1.99]$). This contrast is a *single-annotator, 461-item, post-hoc association*, not an intervention in the experimental sense and not a reference-defect effect: reference replacement simultaneously changes (i) reference content, (ii) expression style/translationese, (iii) BLEU floor level, and (iv) the lexical coupling between retrieval selection and scoring reference. Teacher-forced BT likelihood does not reproduce the probe response under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025).

Scoring against both references jointly leaves the gap at +11.50, *larger* than original: adding the original reference injects a second lexically coupled target. The probe response is thus specific to scoring frames that are lexically coupled to the retrieval query—the original speech-transcribed register under the canonical query, and, more weakly, the human register when the human texts also serve as the retrieval query (§4.1(ii); SI Sup. T). This is a descriptive contrast, not a causal attribution. Under all six reconstruction checkpoints the human-reference gap is negative (-0.15 to -0.65), and the asymmetry is consistent across metrics (chrF 82.2%, BLEU-1 84.0%; SI Table S46). The asymmetric PURE-vs-REC drop (under reference replacement REC drops 50.8%, PURE drops 66.8%, a 1.31× relative-drop ratio; the absolute-drop ratio is 16.68/7.59 = 2.20) is numerically inconsistent with simple symmetric floor compression (SI Table S46), though BLEU's non-linearity precludes a definitive test. The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so semantic adequacy of TN-PURE-v1 is unmeasured.

4.5 Donor-origin contrast

If the released-evaluator probe response requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU vs. REC 12.78 (gap −4.27), a donor-pool substitution estimand of +14.51 BLEU points (SI Table S40). This is a *pool-substitution difference*, not a pure exposure contrast: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor–reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure.

Matched donor-pool analysis (SMD<0.1 control).

Restricting to the 148 items where seen-donor and unseen-donor Jaccard similarities differ by at most 0.05 (SMD 0.065 vs. +0.554 on the full 641), the pool-origin contrast is +9.56 BLEU [+7.47, +11.76]. The change from the unmatched +14.51 is not a percentage attribution—the two estimates target different populations and the caliper shifts length, density, and difficulty distributions. This is a Jaccard-balanced association, not an identified training-exposure effect (pool size and donor diversity remain uncontrolled; `results/matched_donor_pool.json`, SI Sup. E).

Pool-size-controlled resampling.

Equalling pool size directly—repeatedly subsampling 640-item donor pools from the 7,060-item training pool, applying the same retrieval rule, and scoring the released evaluator's cached hypotheses—gives, over 1,000 resamples, a pool-origin contrast of +8.38 BLEU (95% conditional simulation interval [+7.44, +9.34], fraction positive 1.00; `results/donor_pool_resampling.json`). Pool size and donor diversity account for part of the unmatched +14.51, but a substantial contrast persists at equal pool size; it remains an association, not an identified exposure effect.

Balanced factorial and path decomposition (SI).

A 605-query factorial matching seen/unseen donors at high/low LaBSE similarity decomposes the unmatched contrast into a pool-origin main effect of +2.08 [+1.56, +2.57], a similarity effect of +6.96 [+5.93, +8.08], and an interaction of +3.70 [2.63, +4.79] (Holm p=0.0003); across reconstructions the seen main effect shrinks to +0.51. Because seen donors remain more similar than unseen within strata (SMD +0.4), this is a provenance contrast, not an identified training-exposure effect and not a balanced design; the full decomposition, its target population, and its relation to the caliper-restricted and resampling analyses (three different target populations) are reported in SI Sup. E.

4.6 Competitive-system context

Because the audit's adversarial probe is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: the "Ground Truth" row reproduces our canonical REC numbers exactly (BLEU-4 12.78, Walsh et al. (2025); SI Table S35). The retrieval-based winner USTC-MoE sits roughly even with REC on lexical BT metrics (BLEU-4 -0.72; SI Table S35). We re-decoded two realistic retrieval outputs under all available evaluators (SI Sup. P): the deployment-realistic noisy-gloss retrieval pipeline decodes at 12.65 under the released evaluator (gap -0.13 vs. REC), and our rebuild of the winner's fixed outputs from its released intermediates decodes at 2.42 (gap -9.94; a failed reproduction, not representative of the original entry). Under all 69 unique non-degenerate reconstructed evaluators of the canonical (clip=1.0) panel—70 cross-evaluation decodes, the byte-identical `cf_202/ls_202` pair counted once—including the eight faithful runs, the noisy-gloss system gives negative gaps throughout (range $[-3.22, -0.07]$; faithful sub-range $[-3.22, -1.59]$), and USTC-MoE remains deeply negative ($[-9.52, -4.28]$). Neither realistic system recovers a retrieval-favoring ordering under any canonical evaluator; cross-evaluating these systems under the unclipped sensitivity evaluators of §4.3 is left to follow-up work. The $+10.24$ gap is specific to the constructed adversarial probe, and we do not claim that real competition rankings would change across checkpoints. Archiving fixed per-team outputs alongside public scores is the most direct suggestion we can offer for enabling rank-stability audits.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor–reference Levenshtein distance; removing the signer block changes R^2 by $\leq 1\%$ (ΔR^2). The response variable is a sentence-level smoothed-BLEU proxy: the per-item PURE-minus-REC difference computed by aggregating each item's n-gram sufficient statistics under the corpus exp-smoothing convention. This proxy is *not* an additive decomposition of the non-linear corpus BLEU (the mean of the per-item proxies does not equal the corpus difference), so the regression explains the sentence-level proxy response, not exact corpus contributions; SEs are heteroskedasticity-robust (HC3) and also reported with two-way clustering on signer and broadcast show as a sensitivity check. The model reported is the AIC-minimal specification across a small grid of additive and pairwise-interaction candidate regressors selected before fitting; we do not claim it is the unique correct model. Full coefficient table, alternative-model sensitivity, and figure in SI Sup. K.

5 Discussion

The audit yields a sharpened picture of what the SLRTP2025 public artifacts reproduce, organised around RQ1 (reconstruction), RQ2 (reference-frame and donor-origin sensitivity), and one descriptive observation (probe non-reproduction and its resolution).

RQ1—Competence recovered first, the signature second; both were our defects. After seven implementation defects were identified and corrected (§3.4), the faithful family reaches dev BLEU-4 11.59–12.62 against the released 13.38, with test-split decoding up to 13.11 against the released 12.78. The earlier "failed reconstruction" competence gap is therefore attributable predominantly to implementation defects in our own reconstruction code—a result that itself illustrates how easily an eval-only release invites exactly such defects (four of the seven were invisible without forensic arithmetic against the released training log). The training-pool signature initially appeared to be the genuine residual: full-pool readout 78.8 BLEU / 70.7% exact match against a constructible ceiling of 12.08 / 2.5% across 69 runs. The clipping sensitivity (§4.3) closes it as well: unclipped training—the framework default the released config implies—reproduces the signature (99.98 / 99.9%) and probe response (+11.19) at dev 16.32, so the eighth defect was not an artifact gap but a wrong hyperparameter inference whose effect was invisible in every training-time metric. What remains unreproduced is the exact released operating point (readout 78.8 at dev 13.38): the clip ladder brackets it—readout between clip 3.0 and 5.0, gap between clip 5.0 and no clipping, dev between clip 1.0 and 2.0—but no single threshold matches all coordinates; the residual unobserved dev interval $(12.62, 13.38)$ of the canonical panel is likewise exceeded, not filled, by the sensitivity runs.

Descriptive observation—probe non-reproduction was real, uniform, and wrong as an attribution. All 69 non-degenerate constructible evaluators of the canonical (clip=1.0) panel give negative probe gaps (range $[-2.80, -0.21]$), robustly across seeds, families, and partially corrected protocols. Because the faithful runs match or exceed the released evaluator's test-split decoding competence while still lacking the signature, the non-reproduction could not be attributed to surface competence—yet attributing it to an unrecovered training mechanism was also wrong: one hyperparameter, shared silently by all 69 runs, was responsible. Two morals follow. First, uniformity across many runs is not evidence of artifact insufficiency when the runs share an untested inference; the 69-run panel was a robustness study of a wrong prior. Second, the probe response and the memorisation signature are jointly constructible properties of the released recipe, not mysteries: they co-occur in the unclipped regime and in the released evaluator, and the within-regime readout–gap association ($\rho = -0.56$ among clip=1.0 checkpoints) reverses sign across the clip ladder.

RQ2—Reference-frame and donor-origin sensitivity. On the matched 461-item confidence=1 subset of Czehmann et al.'s Czehmann et al. (2026) human back-translations, reference replacement is associated with a gap reduction to +0.95 from +10.03—a single-annotator reference-frame contrast that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects. The matched donor-pool analysis (§4.5) shows the pool-origin association persists at +9.56 under SMD<0.1 Jaccard matching; the pool-size-controlled resampling (640-vs-640, §4.5) gives +8.38 (conditional simulation interval $[+7.44, +9.34]$). All three pool-origin estimates remain associations, not identified training-exposure effects.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-artifact reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025). Chen et al.'s Chen et al. (2022) reproduction failure of a released metric had the same anatomy we document here for a full evaluator: an eval-only release, underspecified training semantics, and reconstruction attempts that failed until the semantics were pinned by forensic comparison with the released artifacts.

Scope limitation: our reconstruction vs. public-artifact sufficiency.

Our conclusion is strictly about what our disclosed implementations construct from the public artifacts. After the eighth correction, the artifacts reproduce—within our implementation—surface competence, the training-pool signature, and the probe response; three caveats bound the sufficiency claim. First, the exact released operating point (readout 78.8 at dev 13.38) is bracketed by the clip ladder, not matched, and the CTC-term normalisation remains unidentifiable from the released log, so residual quantitative differences of unknown sign persist. Second, the clipping resolution rests on source semantics of the signjoey framework (absent config fields $\Rightarrow$ no clipping), an inference about the released run's framework version rather than a released training transcript. Third, all constructions share our own data pipeline; an independent implementation could still diverge. The earlier, stronger negative claim ("the artifacts do not reproduce the signature") is withdrawn with the disclosure of the eighth defect.

Value, finding, and conclusion reproducibility.

Following the tripartite framework of Cohen et al. (2018): *Value* (recomputing the headline numbers from the released artifacts) is reproduced—REC 12.78, PURE 23.02, gap +10.24 regenerate from the canonical donor registry. *Findings* are now reproduced across all three dimensions: surface competence (faithful family 11.59–12.62 dev), the training-pool readout signature, and the probe response (both reproduced unclipped, §4.3), with the exact released operating point bracketed but not matched. *Conclusions* about artifact sufficiency hold in a stronger operational sense than in earlier rounds of this audit, subject to the three caveats of §5.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: an undocumented framework default can masquerade as an unreproducible property. The audit's sharpest result is the eighth defect: because the released `config.yaml` specifies no clip fields, the underlying framework's default (no gradient clipping) silently governs whether the evaluator memorises its training pool and responds to the replay probe—and nothing in the released log distinguishes the regimes, since dev BLEU, NLL, and CTC all look healthy at clip 1.0. Sixty-nine runs sharing our wrong 1.0 inference uniformly hid both properties. Benchmarks should release the exact training command and seeds, pipeline scripts beyond the manifest, a SHA-256 manifest of training tensors, intermediate checkpoints, and the fixed per-team pose outputs; our seven earlier defects—several pin-able only by arithmetic against the released training log—show that even the released numbers under-determine the code that produced them.

Lesson 2: on template-dense benchmarks, retrieval entries may benefit from donor-exclusion disclosure. A donor-exclusion sweep is cheap and exposes the reference-coupling gap. The training-pool readout ratio (released ≈ 5.9; constructible clip-1.0 family 0.93–1.55; unclipped ≈ 6.2) and input-side pose permutation with output-equivariance verification (SI Sup. M) are two informative descriptive probes; generalisation to other benchmarks is untested.

Lesson 3: reference-coupling artifacts warrant a supplementary reference check. The reference-frame contrast under Czehmann et al. Czehmann et al. (2026) back-translations is a single-annotator case study and should not be generalised beyond this signer. Template-dense benchmarks could adopt *multi-annotator* human back-translation resources as a supplementary reference set and flag systems whose gap shifts between frames; with a single annotator, direction is informative but magnitude is not.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). Per-team pose outputs are not archived; our cross-evaluator studies decode at or below REC under all 69 canonical non-degenerate evaluators (all clip 1.0). The faithful family terminates at the plateau scheduler's numerical LR floor (4.4e-8) rather than the configured floor (1e-8), with best checkpoints unchanged for the final 218–338 validations; the released run itself ended mid-schedule at 101 epochs, so the two stop points are not identical—§4.3 evaluates checkpoints fixed at the released run's best and final steps and finds the conclusions unchanged. The clipping resolution rests on signjoey source semantics for absent config fields and is confirmed behaviourally, not by a released training transcript; the CTC-term normalisation remains unidentifiable from public artifacts, and the sensitivity grid of §4.3 brackets both. The unclipped reconstruction overshoots the released operating point (readout 99.98 vs. 78.8; dev 16.32 vs. 13.38), so the exact released configuration is bracketed, not recovered; the unclipped variant is confirmed across three seeds (gaps +11.19/+11.31/+10.31), the intermediate clip thresholds and the signjoey-exact run are single-seed. The human back-translations are a single-annotator resource without second-annotator adjudication; the reference-frame contrast is conditional on one signer's interpretation (qualitative direction stable across confidence tiers; SI Table S41). The probe response is a decoded-BT-text-metric effect, not a sign-language quality judgment; without target-conditioned DGS signer evaluation, semantic adequacy of the donor motion is unmeasured—“evaluator readability” and “signing quality” are distinct constructs, and this audit measures only the former.

6 Conclusion

This audit reports what a best-effort, artifact-conditioned reconstruction of the released SLRTP2025 BT evaluator from public artifacts reproduces, and what stood in the way. Correcting seven implementation defects recovered most of the evaluator's surface competence (dev BLEU-4 11.59–12.62 vs. 13.38; test decoding up to 13.11 vs. 12.78). The two properties that then still failed to reproduce across 69 constructible runs—the training-pool memorisation signature (78.8 BLEU / 70.7% exact match vs. a constructible ceiling of 12.08 / 2.5%) and the adversarial probe response (+10.24 vs. $[-2.80, -0.21]$)—proved to be blocked not by an unrecovered mechanism but by an eighth defect: our gradient-clip threshold of 1.0, imported as an author inference, whereas the released config's absent clip fields mean no clipping under the framework the release derives from. Unclipped, the same recipe reproduces both properties (readout 99.98 BLEU / 99.9% exact match; probe gap +11.19) at dev 16.32, with the released operating point bracketed by the clip ladder; the readout–gap association is regime-dependent (negative among clipped checkpoints, positive across clipping regimes). The probe itself behaves as an adversarial unit test: confirmed on a never-explored split (+8.10), absent under every text-free donor-selection rule tested (reference-free pose retrieval collapses to -12.33), present only on the coupled diagonal of the retrieval-query $\times$ scoring-frame grid, and absent for deployment-realistic systems. The audit's lasting contributions are the forensic method (pinning released-config semantics by arithmetic against the released training log, and framework source defaults against config absences), the provenance, checkpoint-registry, and evaluator-audit infrastructure, the dimension-by-dimension reproducibility accounting, and the cautionary mechanism it documents: an undocumented framework default can make a reproducible property look unreproducible through sixty-nine robustly negative runs.

Statements and Declarations

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Conflict of interest.

The authors declare no competing interests.

Data governance and ethics.

The datasets and model weights are available under research-specific terms: PHOENIX-2014T under its RWTH Aachen academic licence; the SLRTP2025 pose bundle and released BT evaluator under challenge terms; the Czehmann et al. Czehmann et al. (2026) back-translations under CC BY-NC-SA 4.0; CSL-Daily under provider-specific academic terms. The artifact repository is a licence composition: the root MIT Licence covers only the RCP audit software; per-directory

LICENSE/NOTICE files document the remainder (including ShareAlike obligations for the Czehmann CSVs). Pose tensors retain signer identity and biometric features; we release per-item decoded text (low re-identification risk) and a curated subset of trained checkpoints, but not per-signer raw poses. The original consent forms are not publicly available, so we cannot independently confirm that informant consent covers derivative replay-probe analysis; no new human-subjects data was collected. No Deaf/DGS stakeholders were involved in formulating the research questions or interpreting results: the adversarial-probe framing, reference-frame analysis, and recommendations reflect a hearing-researcher perspective on a DGS resource, and Deaf-led sign-language AI research norms should guide any follow-up Desai et al. (2024). The Czehmann et al. back-translations we build upon were produced by a Deaf fluent L2 DGS signer.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md. The artifact is version-controlled (git commit `a2741b1`, 2026-08-18); a claim-to-file manifest mapping each headline number to its source file and verification command is at `claim_manifest.json` (artifact root), and a single-command consistency check (`make check-consistency`, invoking `scripts/check_paper_consistency.py`) verifies the paper's counts against the disk-scanned registry. The L1 audit environment is captured by `requirements-audit.txt` (4 packages: numpy, scipy, sacrebleu, pyyaml; install via `make install-audit`); the full `requirements.lock.txt` (236 packages) is needed only for L3 re-decoding and L4 from-scratch retraining. A persistent DOI-bearing Zenodo release is deposited at https://doi.org/10.5281/zenodo.21987786, available at submission and not deferred to acceptance. The supplementary material (`supplementary.pdf`) contains all SI appendices. Upstream datasets are available under research-specific terms documented in the artifact README and per-directory LICENSE/NOTICE files (PHOENIX-2014T under RWTH Aachen academic licence; Czehmann et al. human back-translations under CC BY-NC-SA 4.0, redistributed with attribution under those terms; CSL-Daily and SLRTP2025 under provider-specific academic terms).

Author contribution.

Both authors contributed to the study conception, design, data analysis, and manuscript preparation, and both read and approved the final manuscript. Specific contributions (CRediT taxonomy): **Xin Ji**: conceptualisation, methodology, software, investigation, formal analysis, data curation, writing—original draft, visualisation. **Cheng Zhang**: methodology, supervision, writing—review and editing, project administration.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact. Each appendix is annotated with the question it answers. **Sup. A** Missing Test ID Sensitivity; **Sup. B** Per-seed Legacy and Extension Cells; **Sup. C** Analysis Provenance and Multiplicity; **Sup. D** Canonical Checkpoint Registry; **Sup. E** Donor-Pool Substitution Decomposition; **Sup. F** Config-Faithful and Perturbation Details (incl. the four-column config-mapping table); **Sup. G** Cluster-Aware Bootstrap; **Sup. H** Supplementary Oracle Diagnostics; **Sup. I** Template-Density, Holdout Boundary, Donor-Exclusion; **Sup. J** Transcript-Slot Diagnostic; **Sup. K** Competence Extrapolation and Per-Item Gap Decomposition; **Sup. L** CSL-Daily Boundary Control; **Sup. M** Leakage Sanity Checks; **Sup. N** Pre-Exclusion Materialization Sensitivity; **Sup. O** BLEURT-Substitute Validation; **Sup. P** Main-Text Tables (moved from the main text); **Sup. Q** Additional Protocol Details; **Sup. R** Reconstruction Sanity Checks; **Sup. S** Frozen-Split Confirmation and Reference-Free Retrieval (incl. the full probe-multiplicity matrix); **Sup. T** Retrieval-Query Decoupling Grid; **Sup. U** Gradient-Clipping and CTC-Normalisation Sensitivity. Each appendix is annotated with the question it answers; labels in `supplementary.pdf` match the references throughout the main text.

Full checkpoint registry in SI Sup. D.

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